Project Proposal: Rebuilding and Dissecting U-Net

CS-320 Artificial Neural Networks & Deep Learning, Fall 2026 | BS-AI 2024F, NBC, NUST

Group 5 | Instructor: Dr. Zirak Khan

Project Overview

We will rebuild U-Net (Ronneberger, Fischer & Brox, MICCAI 2015) from scratch in PyTorch. U-Net labels every pixel of an image. An encoder shrinks the image to capture *what* is in it, a decoder grows it back to say *where*, and skip connections carry fine detail across. Its elastic-deformation augmentation lets it learn from very few images (30 in the original task). We will reproduce the paper's key experiment at Colab scale, run the model on a new dataset, and study it through one-factor-at-a-time ablations. **Datasets:** ISBI 2012 cell segmentation (reproduction) and Oxford-IIIT Pet trimaps (new dataset). Code is in our GitHub repository, with all members as collaborators.

Team and Roles

Member	Role	Responsibilities and own component
Muhammad Obaidullah - 513507	Paper lead	Architecture diagram with tensor shapes, hand parameter count, reproduction. Implements the encoder–decoder blocks and skip connections; ablates skip connections and depth.
Inshan Haider - 502067	Data lead	Datasets, DataLoaders, shape checks, normalisation, evaluation. Implements elastic-deformation augmentation and Dice/IoU metrics; ablates augmentation.
Shayan Afzal - 513369	Experiments lead	Ablation table, Gradio demo, report, error analysis. Implements the loss functions and training loop; ablates loss (BCE vs Dice) and base channels.

Problem Statement

Pixel-level segmentation normally needs large labelled datasets, which are costly in biomedical imaging. U-Net claims accurate segmentation from very few images, but it is unclear how much of that comes from the skip connections, the depth, the loss, or the augmentation, and whether the result carries over to a different kind of image. We will reproduce the result, test it on a new dataset against a simple baseline, and measure which design choices matter most.

Research Questions

RQ1. Can we reproduce U-Net's segmentation quality on ISBI 2012 at Colab scale (Dice, IoU), and how large is the gap to the paper?

RQ2. Does U-Net still outperform a simple baseline on a new dataset (Oxford-IIIT Pet), where images and classes differ from the original?

RQ3. Which one-factor change (skip connections, depth, base channels, loss, augmentation) affects performance most?